

```
#include<stdio.h>
#define MAX 3
int oqueue[MAX];
int front=-1, rear=-1;
void insert()
{
    int value;
    if((front==0&&rear==MAX-1)||front==rear+1)
    {
        printf("Circular Queue Overflow!\n");
        return;
    }
    printf("Enter element to insert:");
    scanf("%d",&value);
    if(front==--1)
        front=rear=0;
    else if(rear==MAX-1)
        rear=0;
    else
        rear++;
    oqueue[rear]=value;
    printf("%d inserted into the circular queue.\n",value);
}
void delete()
{
    if(front==--1)
    {
        printf("Circular Queue Underflow!\n");
        return;
    }
    printf("%d deleted from the circular queue.\n",oqueue[front]);
    if(front==rear)
        front=rear=-1;
    else if(front==MAX-1)
        front=0;
    else
        front++;
}
void display()
{
    if(front==--1)
    {
        printf("Circular Queue is empty.\n");
        return;
    }
    printf("Circular Queue elements:");
    if(front<=rear)
    {
        for(int i=front;i<=rear;i++)
        {
            printf("%d ",oqueue[i]);
        }
    }
}
```

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```
32     if(front==rear)
33         front=rear=-1;
34     else if(front==MAX-1)
35         front=0;
36     else
37         front++;
38 }
39 void display()
40 {
41     if(front==--1)
42     {
43         printf("Circular Queue is empty.\n");
44         return;
45     }
46     printf("Circular Queue elements:");
47     if(front<=rear)
48     {
49         for(int i=front;i<=rear;i++)
50         {
51             printf("%d ",cqueue[i]);
52         }
53     }
54     else
55     {
56         for(int i=front;i<MAX;i++)
57             printf("%d ",cqueue[i]);
58         for(int i=0;i<=rear;i++)
59             printf("%d ",cqueue[i]);
60     }
61     printf("\n");
62 }
63 int main()
64 {
65     int choice;
66     while(1)
67     {
68         printf("\n---Circular Queue Menu---\n");
69         printf("1.Insert\n2.Delete\n3.Display\n4.Exit\n");
70         printf("Enter your choice:");
71         scanf("%d",&choice);
72         switch(choice)
73         {
74             case 1:insert();break;
75             case 2:delete();break;
76             case 3:display();break;
77             case 4:return 0;
78             default:printf("Invalid choice!Try again.\n");
79         }
80     }
81 }
82 }
```

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"C:\Users\Admin\Desktop\18M24CS173\Circular Queue.exe"

---Circular Queue Menu---

1.Insert
2.Delete
3.Display
4.Exit

Enter your choice:2

Circular Queue Underflow!

---Circular Queue Menu---

1.Insert
2.Delete
3.Display
4.Exit

Enter your choice:3

Circular Queue is empty.

---Circular Queue Menu---

1.Insert
2.Delete
3.Display
4.Exit

Enter your choice:1

Enter element to insert:87

87 inserted into the circular queue.

---Circular Queue Menu---

1.Insert
2.Delete
3.Display
4.Exit

Enter your choice:1

Enter element to insert:56

56 inserted into the circular queue.

---Circular Queue Menu---

1.Insert
2.Delete
3.Display
4.Exit

Enter your choice:1

Enter element to insert:34

34 inserted into the circular queue.

---Circular Queue Menu---

1.Insert
2.Delete
3.Display
4.Exit

Enter your choice:1

Circular Queue Overflow!

---Circular Queue Menu---

1.Insert
2.Delete
3.Display
4.Exit

Enter your choice:3

Circular Queue elements:87 56 34

---Circular Queue Menu---

1.Insert
2.Delete

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"C:\Users\Admin\Desktop\18M24CS173\Circular Queue.exe"

```
2.Delete
3.Display
4.Exit
Enter your choice:1
Enter element to insert:34
34 inserted into the circular queue.
```

---Circular Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:1
Circular Queue Overflow!
```

---Circular Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:3
Circular Queue elements:87 56 34
```

---Circular Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:2
87 deleted from the circular queue.
```

---Circular Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:3
Circular Queue elements:56 34
```

---Circular Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:1
Enter element to insert:28
28 inserted into the circular queue.
```

---Circular Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:3
Circular Queue elements:56 34 28
```

---Circular Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:
```

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